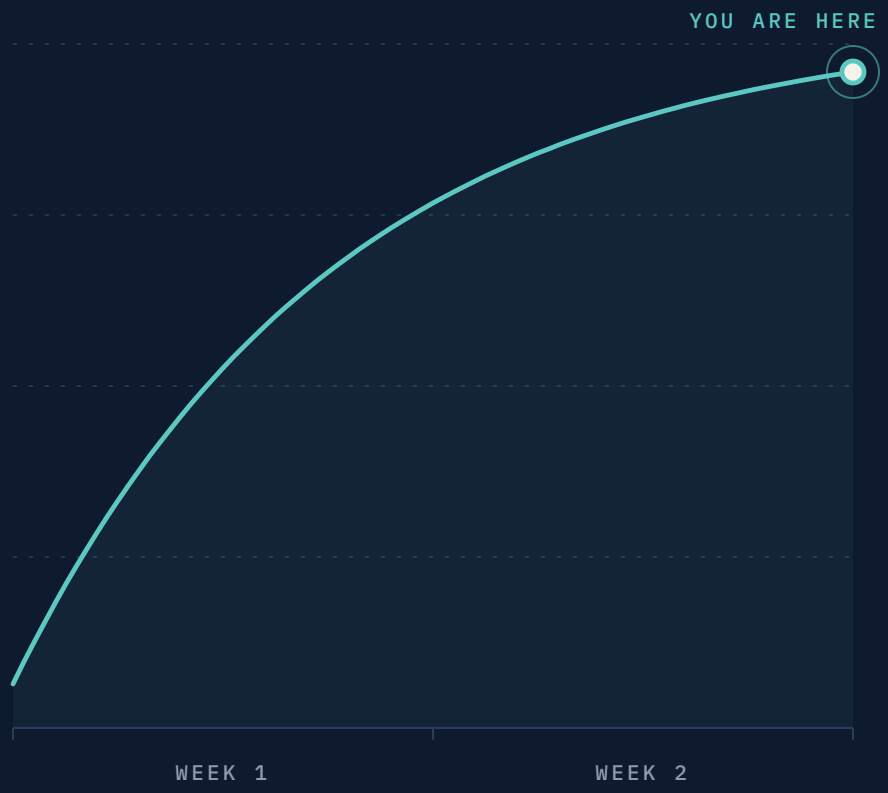
WEEK 2 · DAY 5 · OVERVIEW

Course 5B Capstone



Created by **Jeff Lenamon**

ABOUT 50 MIN WITH EXERCISES / 12 SECTIONS / 2 EXERCISES

This course is educational content created in a personal capacity. Nothing here is investment advice or a recommendation to buy or sell any security. All portfolio data is synthetic.

Problem Statement

BUSINESS CONTEXT

A colleague hands you a copy of the 2 Yr and 10 Yr yields and a synthetic HY spread, in bp, with blank days and a few wrong prints.

From it you build the 2s10s slope and the HY, and ask the course's two questions of each.

Does any model of the change get closer than the last value carried forward?

Does any rule for the size of a change get closer than a constant?

Every output is a value for a date inside the file, from the dates before it, never a forecast of rates or spreads.

The work is yours: each question has an empty cell and a **check()** line.

OBJECTIVES

Repair a dated copy with a log, and test and describe both series.

Score models of the change and of its size over Day 2's folds, and choose by a rule written first.

Open the held-out years once, score every model there, and say where the models miss.

ABOUT 50 MIN WITH EXERCISES / 12 SECTIONS / 2 EXERCISES

Your Route Through Day 5

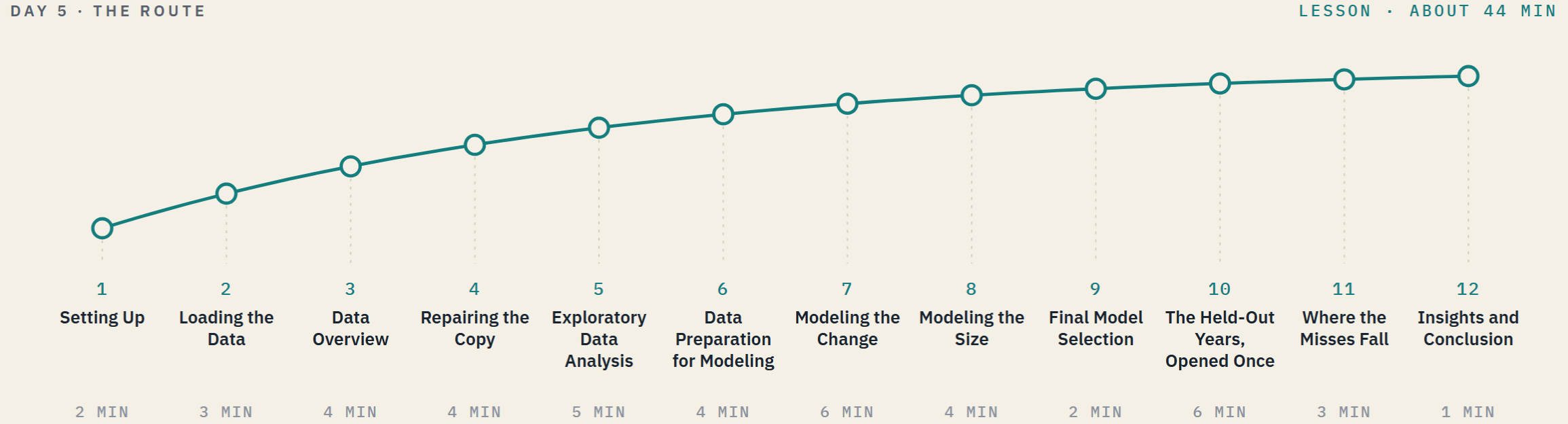


FIGURE 2 Twelve sections, in order, with about how long each takes.

Time. About 44 minutes for the lesson, then about 6 for the two exercises.

Fix the seed and load the hashed checker.

Setting Up

THE SEED AND THREADS

```
os.environ["OMP_NUM_THREADS"] = "1"
SEED = 42
```

THE ANSWER CHECKER

```
import hashlib
def check(label, answer, code, places=None):
    if answer is ...:
        print(label + ": not attempted yet")
        return
    if places is not None:
        answer = round(float(answer), places)
    text = label + "|" + str(answer).strip().lower()
    if hashlib.sha256(text.encode()).hexdigest()[:12] == code:
        print(label + ": correct")
    else:
        print(label + ": not yet. You have " + str(answer))
```

Read both files and make the planted copy.

Loading the Data

READ BOTH FILES AND CUT

```
yields = pd.read_csv(DATA + "treasury_par_yields.csv", parse_dates=["date"], index_col="date")
spreads = pd.read_csv(DATA + "course3_synthetic_spreads.csv", parse_dates=["date"], index_col="date")
CUT = "2022-12-30"
print("Development:", len(development), "dates | held out:", (yields.index > CUT).sum(), "dates, closed for now")
```

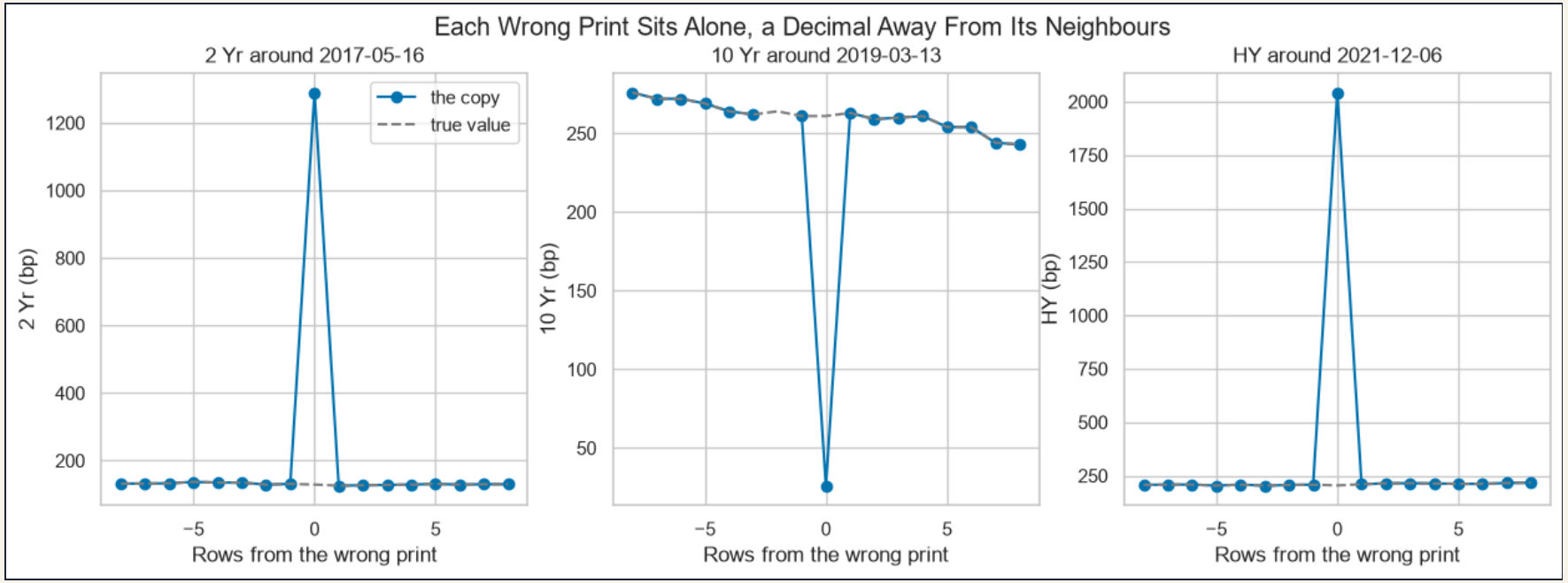
```
Development: 2001 dates | held out: 939 dates, closed for now
```

THE SECOND FEED

```
blank_first = {column: published.index[published[column].isna()][0] for column in published.columns}
feed_rows = []
for column in published.columns:
    for date in (wrong_dates[column], blank_first[column]):
        feed_rows.append((date, column, truth.loc[date, column], "second feed"))
second_feed = pd.DataFrame(feed_rows, columns=["date", "column", "value", "source"])
print("The copy:", published.shape, "| the second feed:", len(second_feed), "cells")
```

```
The copy: (2001, 3) | the second feed: 6 cells
```

Data Overview



QUESTION	WHAT IT ASKS
Blanks	How many cells of **`published`** are blank, and how many rows long is the longest run of blanks in any column?
Wrong Prints	How many cells does it flag in **`published`**, and in **`truth`**?
	The suspect is the row before each flag.

Repairing the Copy

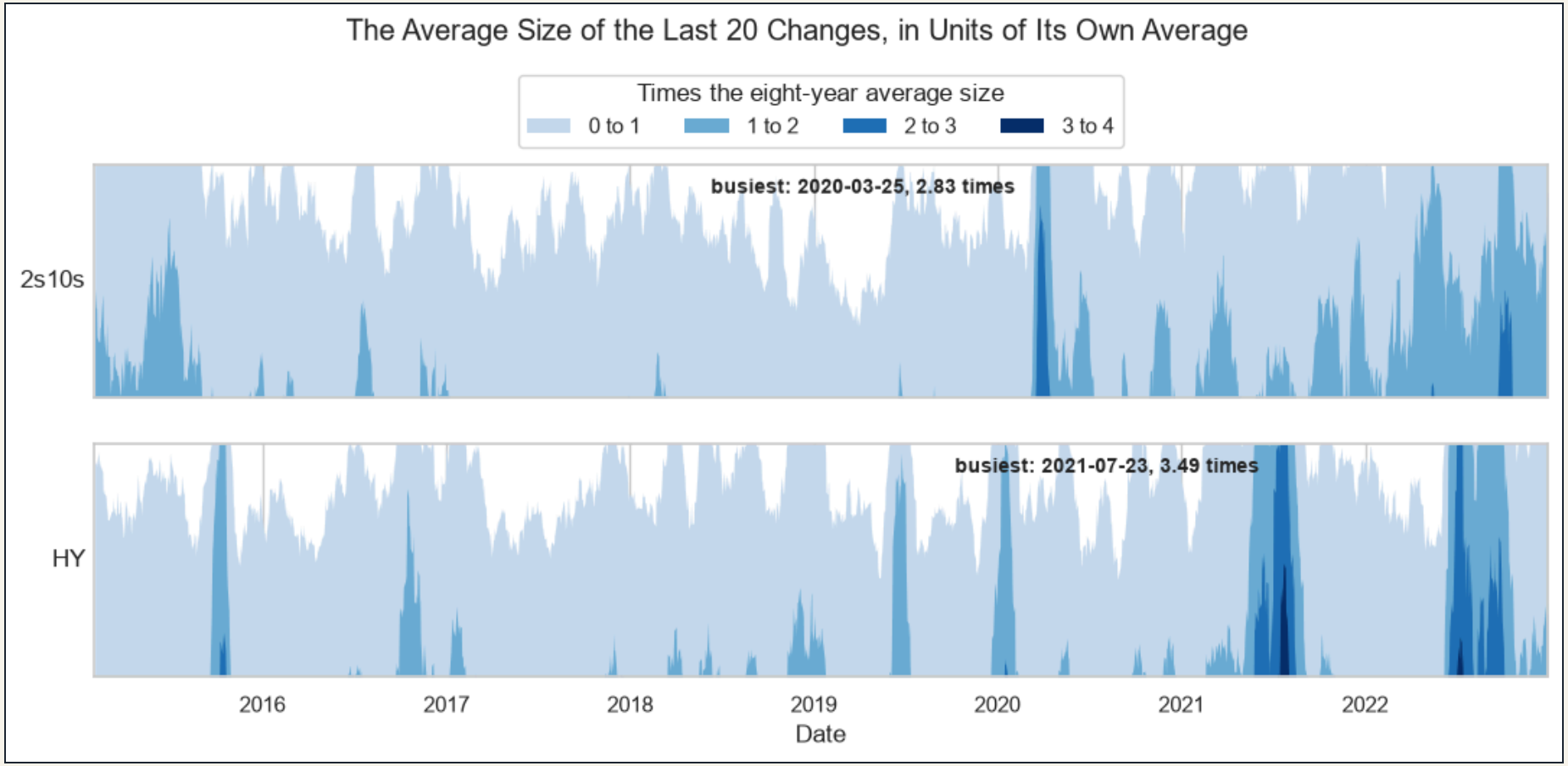
QUESTION	WHAT IT ASKS
The Corrections Table and the Log	Write **`second_feed`** into a copy of **`published`** with Day 1's loop, as **`fixed`**, with the log as **`change_log`**.
Predict 1	After that fill, how many of the 6,003 cells will differ from **`truth`**, and by at most how many bp?
A One-Row Fill	Fill **`fixed`** that way as **`repaired`**.
The Two Series	Build **`levels`**, Day 2's five series, from **`repaired`** (the **`IG`** from **`spreads`**), then **`changes`** and **`SERIES = ["2s10s", "HY"]`**.

WHAT GETS MADE

Made here, in bp: **`truth`** and **`published`** (the true values and the copy), **`second_feed`**, and Day 2's five series from the repaired copy.

> **Every change to the copy is in the log, and the fill is capped at one row.**

Exploratory Data Analysis



The size of a change comes in stretches; the change itself is tested next.

Data Preparation for Modeling

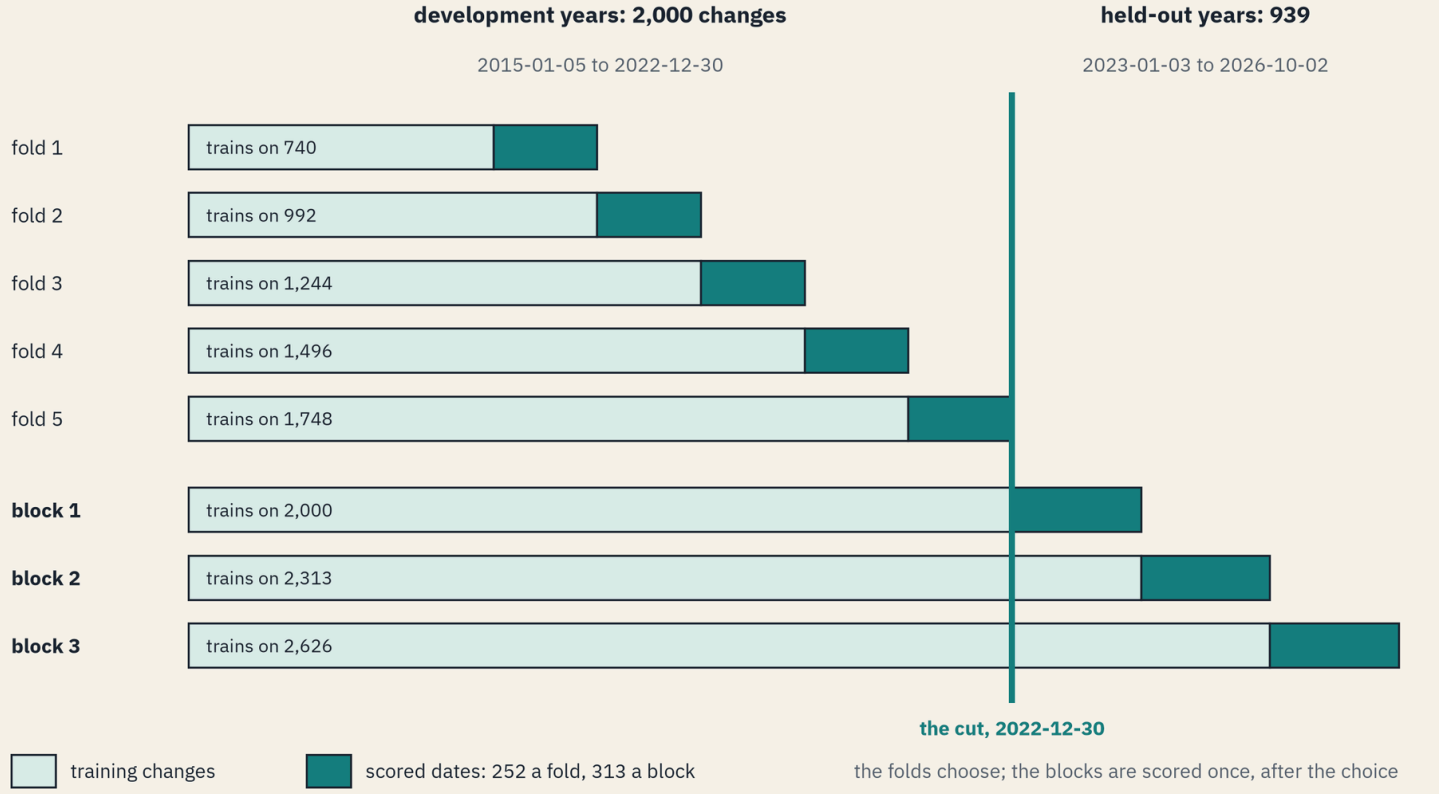


FIGURE 3 2,000 development changes in Day 2's five folds, then 939 held-out changes in three blocks, each refitted.

WHAT THE SECTION ASKS

QUESTION	WHAT IT ASKS
Day 2's Folds and the Bar	Make Day 2's five **`folds`**, and keep the last value's MAE per fold and series in a table **`bar`**.
Lag Features That Pass the Cut Test	Write Day 8's **`lag_features(changes, name)`**, and keep each series' features in a dictionary **`features`**, the first 20 rows dropped once.

Every input uses only earlier dates, and the folds are Day 2's.

Modeling the Change

DAY 2'S LOOP AS ONE FUNCTION

```
def walk_forward(data, splits, make_values, size=False):
    maes, values = [], []
    for fit_rows, test_rows in splits:
        train, test = data.iloc[fit_rows], data.iloc[test_rows]
        values.append(pd.Series(np.asarray(make_values(train, test)), index=test.index))
        maes.append(mean_absolute_error(test.abs() if size else test, values[-1]))
    return pd.Series(maes, index=range(1, len(splits) + 1)), pd.concat(values)
```

QUESTION	WHAT IT ASKS
How Many Terms	Fit **`ARIMA(p, 0, q)`**, p and q from 0 to 2, on fold 1's training changes, a table **`grid`** of BIC.
ARIMA and Smoothing	Score the last value, drift, AR(1), ARMA(1, 1) and smoothing on the folds, a table of fold MAEs per series in **`change_mae`**, and their MASEs in **`mase`**.
LightGBM on Lags	Write **`lgbm_values(train, test, **settings)`** on **`features[train.name]`**, and add LightGBM's defaults and Day 8's setting to **`change_mae`** and **`mase`**.
	Draw each fold's MAE over the last value's, one row per model and series, the mean a diamond.

Modeling the Size

QUESTION	WHAT IT ASKS
Predict 2	What half-life will the same fit give on the repaired 2s10s?
GARCH on the Development Years	Fit Day 7's GARCH(1, 1) on all 2,000 changes of each series, keeping alpha, beta, persistence and half-life in **`garch_table`**.
Three Rules for the Size	Score the three with **`walk_forward(..., size=True)`**: fold MAEs in **`size_mae`**, values in **`size_values`**, a table per series.
	Draw GARCH's value and the EWMA's over the HY's sizes in fold 5, and a Bland-Altman plot of the two.

FROM GARCH TO A SIZE

A GARCH volatility is a standard deviation; for a normal shock the average size is sqrt(2 / pi), about 0.798, times it, so 0.798 times GARCH's volatility is its value for the size (Day 7).

COMPARING TWO RULES

A **Bland-Altman plot** compares two measures of the same thing: each date's difference against their mean, with the mean difference and plus and minus 1.96 standard deviations of it drawn across.

Final Model Selection

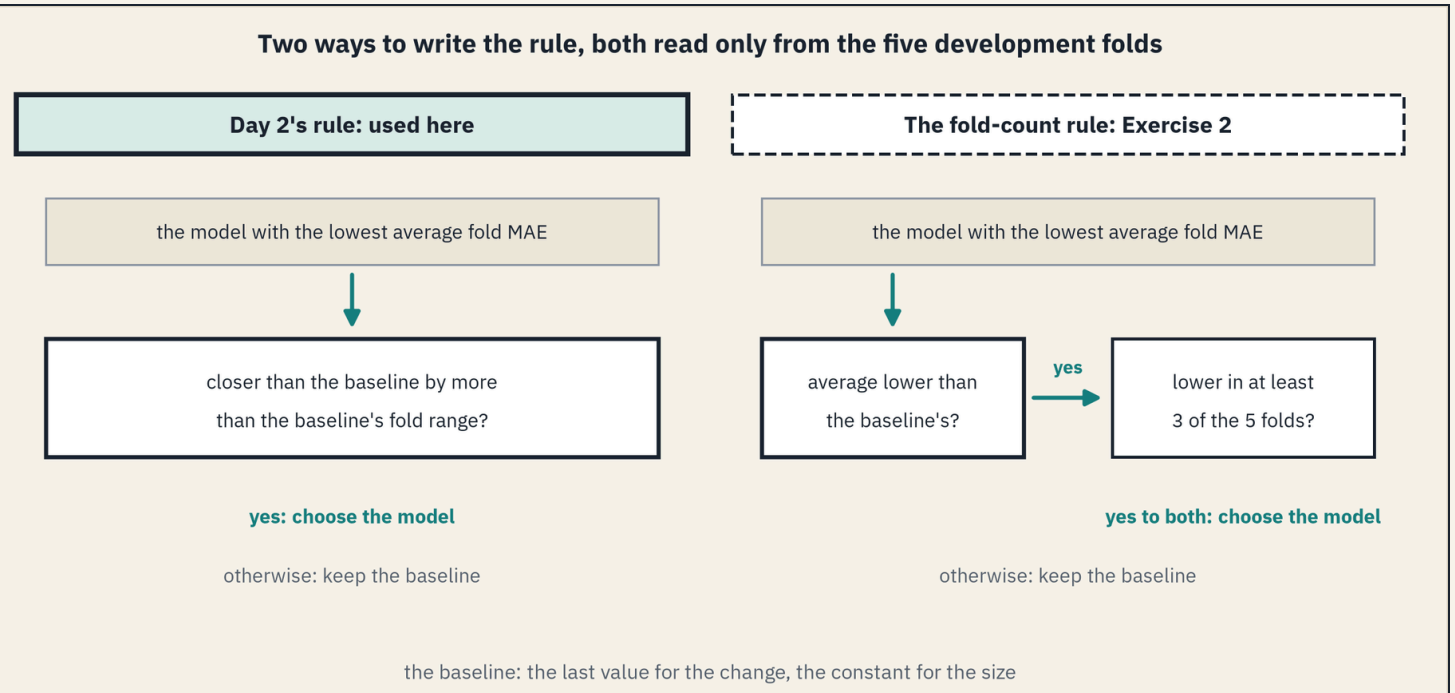


FIGURE 4 Day 2's rule, used here, beside the fold-count rule of Exercise 2, both read from the development folds only.

WHAT THE SECTION ASKS

QUESTION	WHAT IT ASKS
The Rule, Written First	Write Day 2's rule as `choose(fold_mae, baseline)`, and apply it to `change_mae` and `size_mae`, in a dictionary `choice` keyed `("2s10s", "change")` and so on.

The choice is fixed before one held-out value is read.

Score every model once on three refitted blocks.

The Held-Out Years, Opened Once

QUESTION	WHAT IT ASKS
Opening Them	Join **held_levels** under **levels**, make **full_changes** and the **blocks**, and rebuild **features** on **full_changes**.
Predict 3	What will its MAE be on the 939 held-out dates?
The Change, Scored Once	Walk every change model over the blocks: block MAEs in **held_change**, MASEs in **held_mase**.
	Draw each model's MASE on the folds and on the blocks as a dumbbell.
Predict 4	What will that ratio be on the 939 held-out dates?
The Size, Scored Once	Walk the size rules over the blocks: **held_size** and **held_size_values**.
	Draw that curve for GARCH over the constant, 63 dates wide, with the block edges and the development ratio marked.

WHAT THE ONE READ PRINTS

```
939 dates, 2023-01-03 to 2026-10-02 | blank cells: 0
```

Count the tails and the gaps in bp.

Where the Misses Fall

QUESTION	WHAT IT ASKS
Tails GARCH Leaves	Divide each held-out change by it, in a dictionary **`units`**.
	Draw one per series, bins of half a volatility.
	Draw one per series, the dates outside the wedge marked.
The Gaps in bp	Keep each model's held-out MAE less the last value's, in bp, in **`held_gap`**.
	Draw them as bars, with a band of plus and minus the last value's development fold range.

HANGING ROOTOGRAM

A **hanging rootogram** hangs each bin's square-root count from a normal curve's: a bar ending above zero is a bin with too few, below zero too many.

WEDGE PLOT

A **wedge plot** draws each change against its volatility, with rays at plus and minus 3 volatilities: a date outside the wedge is one a normal curve almost never holds.

Insights and Conclusion

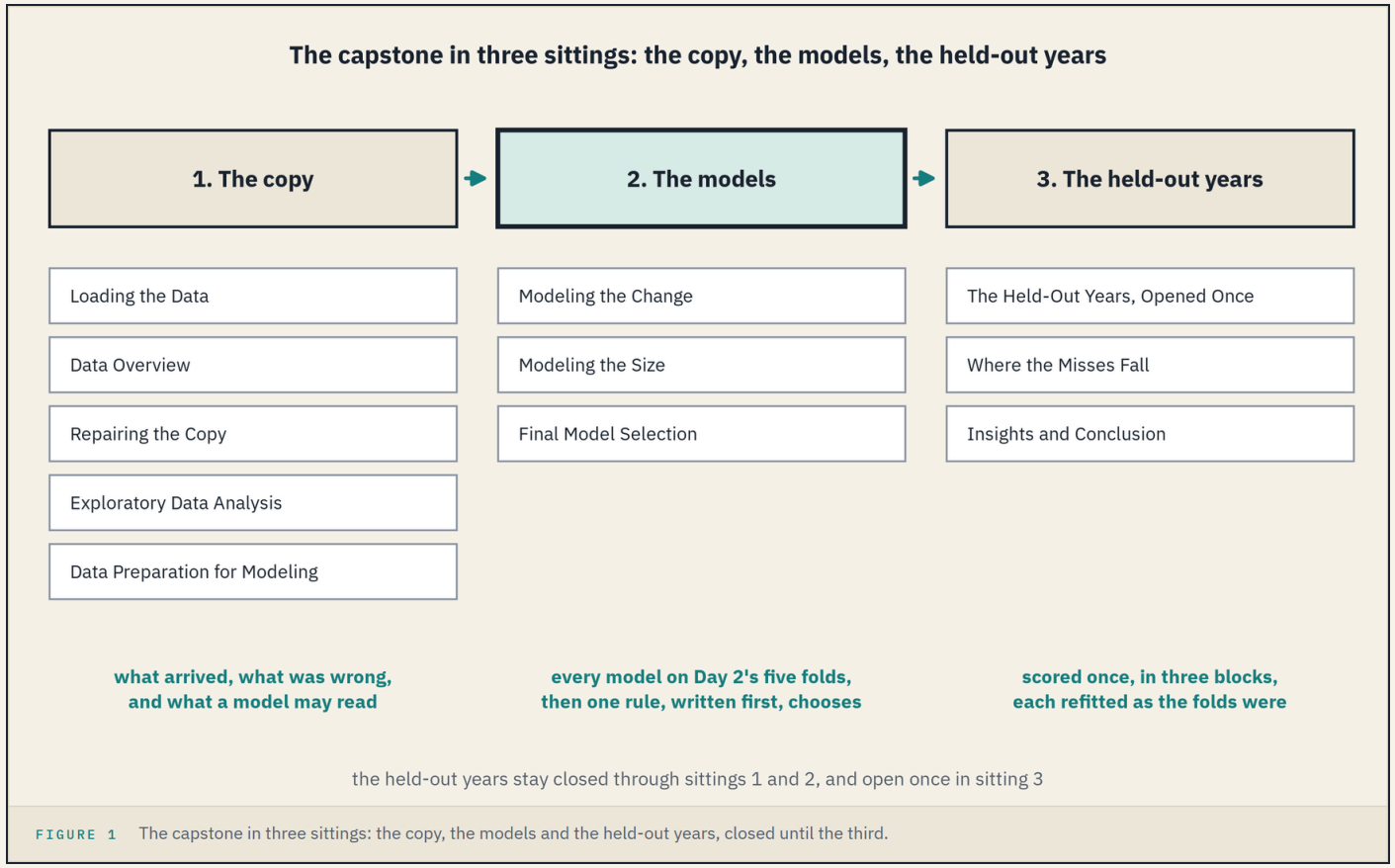


FIGURE 1 The capstone in three sittings: the copy, the models and the held-out years, closed until the third.

WHAT THE SECTION ASKS

QUESTION	WHAT IT ASKS
The Description	Write **`description`**, a few sentences: the repair, the rule, what held on the held-out years and what did not, and what was not tested.

A model describes the dates it was scored on.

You Can Now

- Repair a dated copy with a corrections table, a change log and a capped fill, and check the repair against held-back values.
- Test a series for a unit root and read the memory in its changes and in their size.
- Score models of a change and of its size over walk-forward folds with one function.
- Write a choice rule before scoring held-out dates, and open them once, refitting block by block.
- Draw a Bland-Altman plot, a hanging rootogram and a wedge plot of changes against their volatility.

NEXT

WEEK 1 · DAY 1

Monte Carlo Foundations

Course 5C begins with Monte Carlo: one bond's loss, simulated year by year.